CAUTION: High voltages are present in this laboratory exercise! Do not make or modify any connections with the power on unless otherwise specified! If any danger occurs, immediately press the RED EMERGENCY OFF button on the Three-phase Power Supply Module. Complete this laboratory exercise as quickly as possible to avoid the rise in temperature under load condition. The machines are heavy, make sure to practice safety precautions when moving them from the cabinet to the table top.

Lab 3: Three-phase Salient Pole Synchronous Generator

Activity 3.1: Armature Resistance Measurement

3.1.1. Aim of activity

After completing this exercise, you should be able to measure the armature resistance of a three-phase salient pole synchronous generator.

3.1.2. Equipment required

Table 3.1.1. List of equipment

Qty	Description	Cat. No.
1	Three-phase Salient Pole Synchronous Machine	EM-3330-3A
1	DC Power Supply Module	EN-3310-1A
1	System Transformer	EM-3340-3B
1	Three-phase Power Supply Module	EM-3310-1E
1	Three-pole Current Limit Protection Switch Module	EM-3310-2A
1	Digital DCA Meter	EM-3310-3A
1	Digital DCV Meter	EM-3310-3B
1	Connecting Leads Set	EM-3390-3A
1	Safety Bridging Plugs Set	EM-3390-4A

3.1.3. Schematic

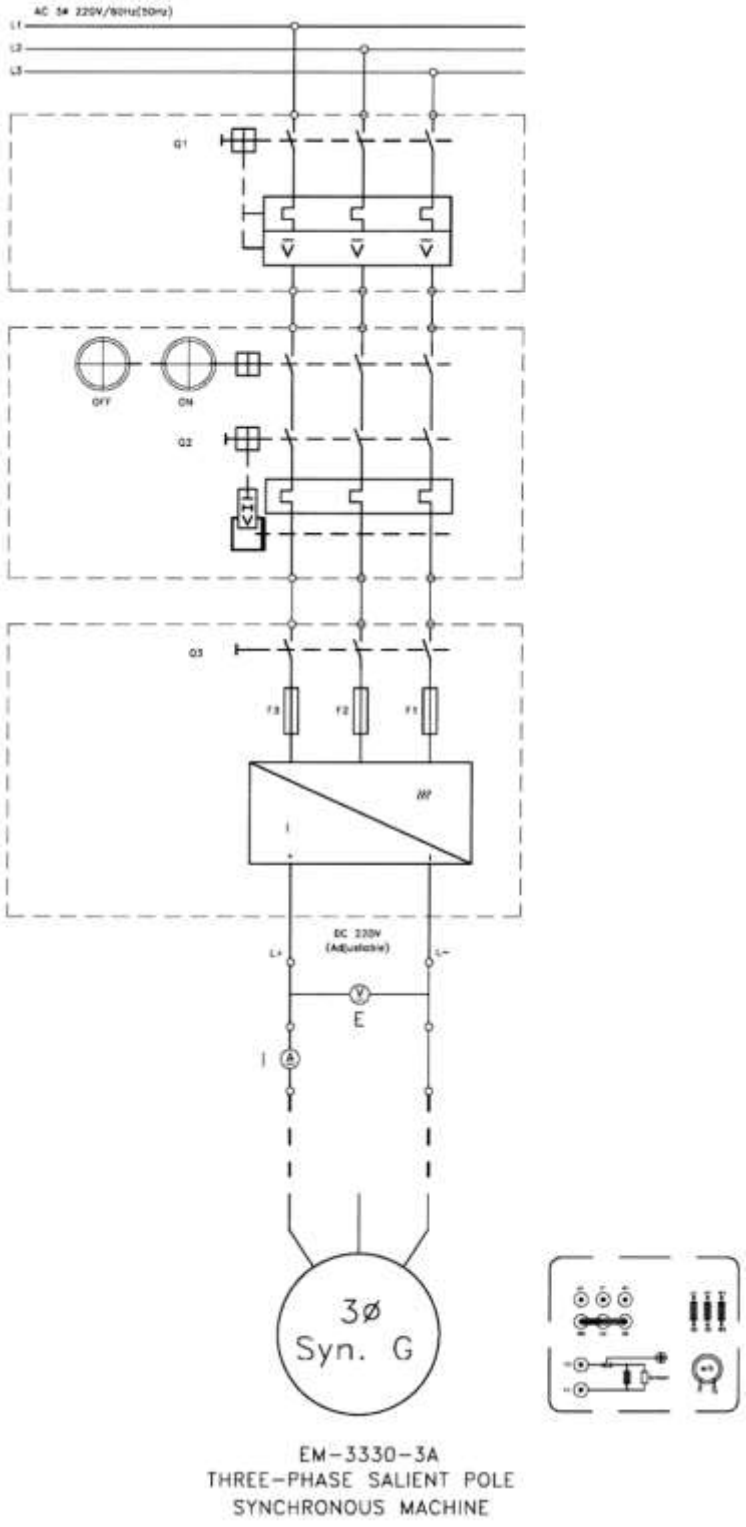


Figure 3.1.1. Circuit diagram for armature resistance measurement

3.1.4. Connection

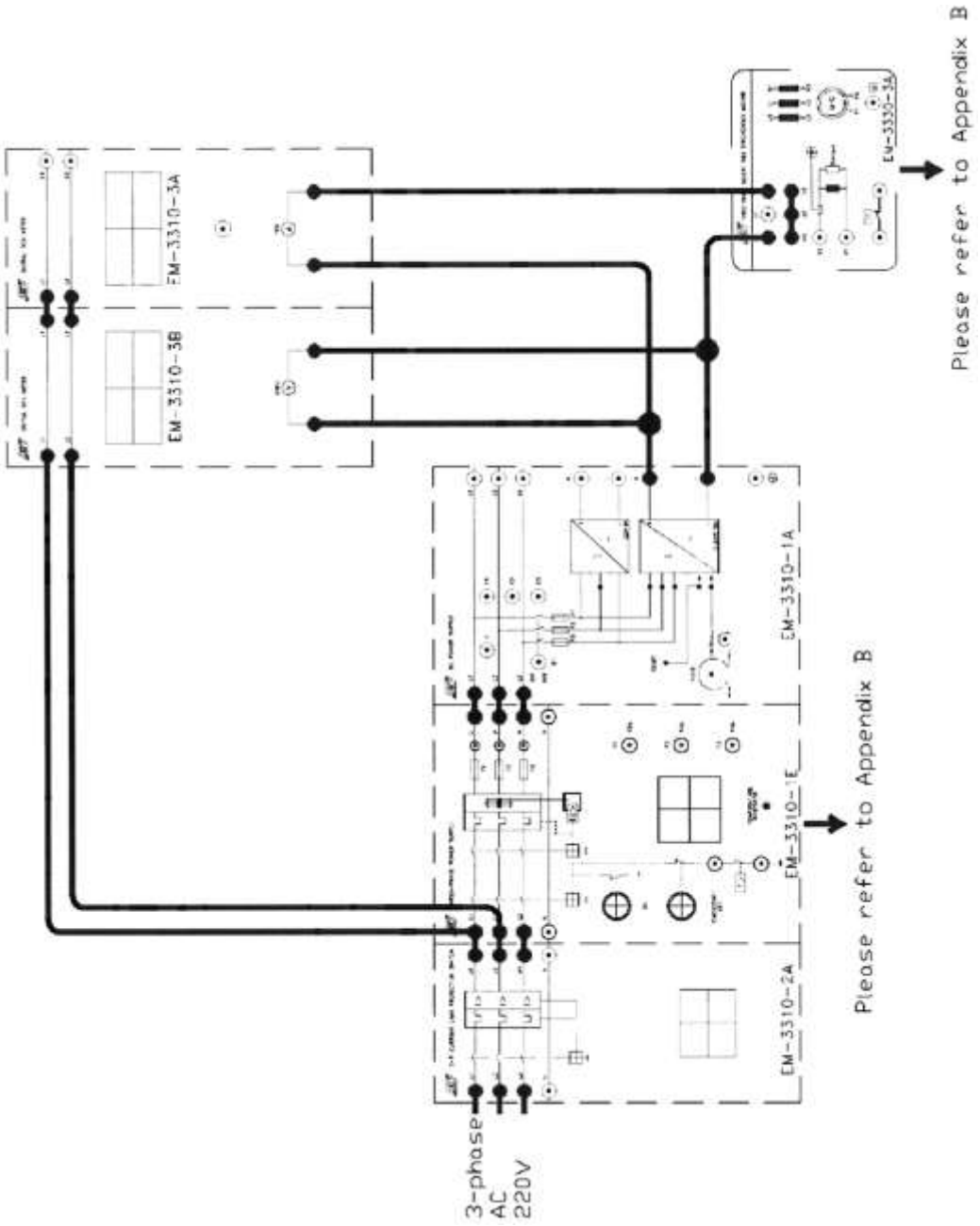


Figure 3.1.2. Connection diagram for armature resistance measurement

3.1.5. Experimental procedure

1. Place the Three-phase Salient Pole Synchronous Machine on the Laboratory Table. Install the required Modules in the Experimental Frame. Construct the circuit in accordance with the circuit diagram and the connection diagram in Figure 3.1.1 and 3.1.2, respectively. Have the instructor or lab technician to check on your completed circuit. The synchronous generator operates in wye.

2. On the DC Power Supply Module, set the V_{adj} knob to the min. position (fully CCW)**.**

3. Sequentially turn on the main switches, System Transformer, 3-P Current Limit Protection Switch, Three-phase Power Supply, and DC Power Supply Modules.

4. Press the START button on DC Power Supply Module. Slowly turn the V_{adj} knob and set the voltage E to 8 V. Record the current I displayed by the Digital DCA Meter in Table 3.1.2. Calculate the armature resistance R_a.

5. Repeat step 4 for other E settings in Table 3.1.2

6. Sequentially turn off the DC Power Supply, Three-phase Power Supply, Current Limit Protection Switch Modules and System Transformer.

3.1.6. Results and discussions

1. Fill in the values obtained from the previous section to Table 3.1.2 below:

Table 3.1.2. Measured values of I, E, P and cos θ

E (V)	I (A)	Calculated R_a (ohm)
8V		
10V		
14V		

2. Discuss and conclude your findings from this activity.

Activity 3.2: No-Load Characteristic

3.2.1. Aim of activity

After completing this exercise, you should be able to demonstrate the operating characteristic of a three-phase salient pole synchronous generator under no-load saturation condition.

3.2.2. Equipment required

Table 3.2.1. List of equipment

Qty	Description	Cat. No.
1	DC Permanent-magnet Machine	EM-3330-1A
1	Three-phase Salient Pole Synchronous Machine	EM-3330-3A
1	DC Power Supply Module	EM-3310-1A
1	System Transformer	EM-3340-3B
1	Three-phase Power Supply Module	EM-3310-1E
1	Synchronous Machine Exciter	EM-3310-1C
1	Three-pole Current Limit Protection Switch Module	EM-3310-2A
1	Digital Power Analysis Meter	EM-3310-3H
2	Digital DCA Meter	EM-3310-3A
2	Digital DCV Meter	EM-3310-3B
1	Digital RPM Meter	EM-3310-3G
1	Coupling	EM-3390-2A
1	Connecting Leads Set	EM-3390-3A
1	Safety Bridging Plugs Set	EM-3390-4A

3.2.3. Schematic

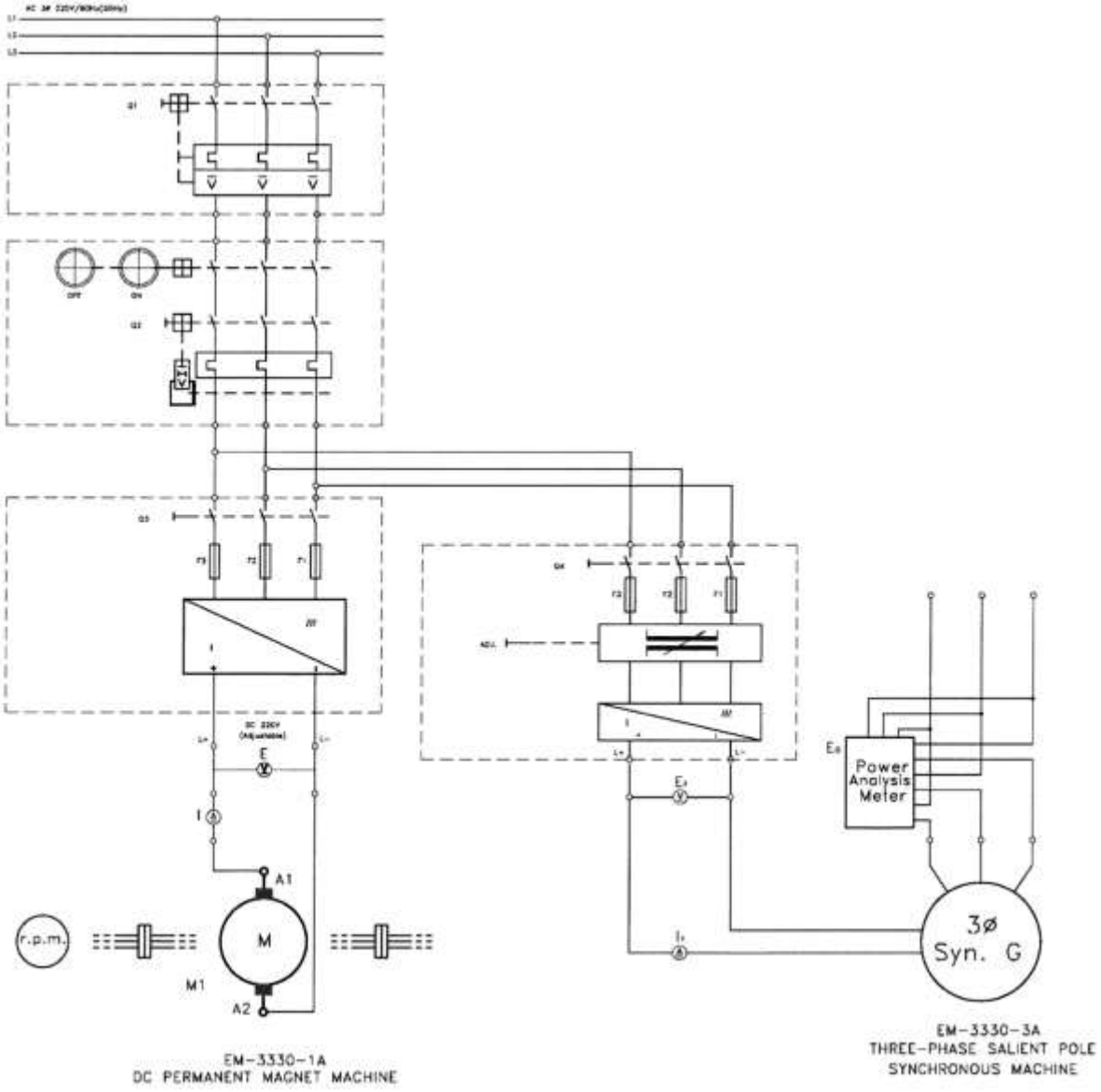


Figure 3.2.1. Circuit diagram for no-load saturation test

3.2.4. Connection

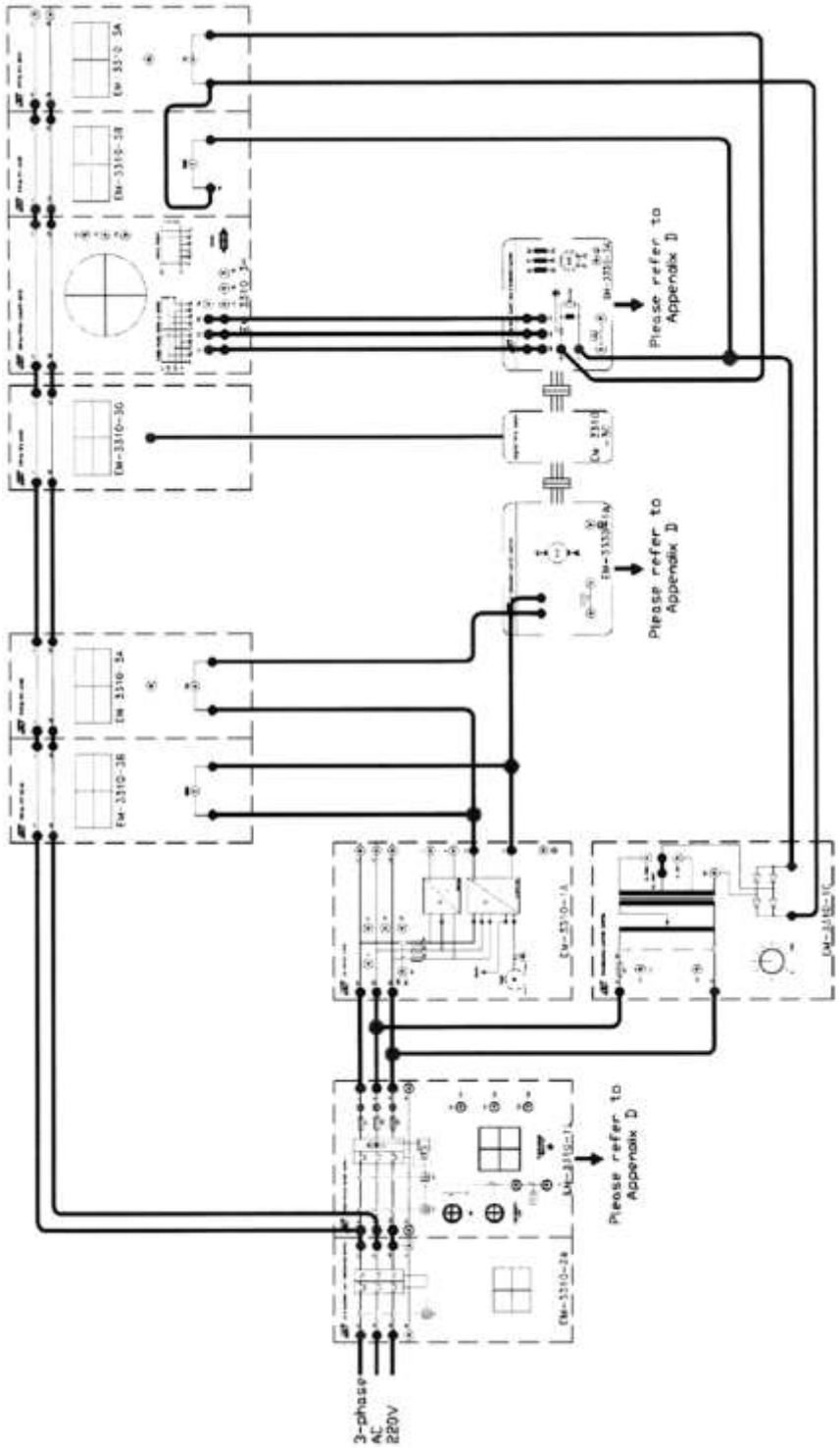


Figure 3.2.2. Connection diagram for no-load saturation test

3.2.5. Experimental procedure

1. Place the DC Permanent-magnet Machine, Three-phase Salient Pole Synchronous Machine, and Digital RPM Meter on the Laboratory Table. Mechanically couple the DC Permanent-Magnet (PM) Machine to the Three-phase Salient Pole Synchronous Machine and the Digital RPM Meter using Couplings. Securely lock the Machine Bases together using the delta screws.

2. Install the required Modules in the Experimental Frame. Construct the circuit in accordance with the circuit diagram and the connection diagram in Figure 3.2.1 and 3.2.2, respectively. Have the instructor or lab technician to check on your completed circuit. The synchronous generator operates in delta.

3. Set the V_{adj} knob on DC Power Supply Module to the min. position. Set the voltage control knob on the Synchronous Machine Exciter Module to the 0 position.

4. Sequentially turn on the main switches, System Transformer, 3-P Current Limit Protection Switch, Three-phase Power Supply, and DC Power Supply Modules.

5. On the DC Power Supply Module, press the START button and slowly torn the V_{adj} knob to increase the motor voltage until the generator rotates at its rated speed. Maintain the speed through this exercise. **Note: The rated speed of the generator is 1,500 rpm for 50-Hz power.**

6. Turn on the Synchronous Machine Exciter.

7. On the Synchronous Machine Exciter Module, slowly turn the voltage control knob so that the field current I_f (obtained from the Digital DCA Meter) is 0 A. Record the generator output voltage Eo (obtained from the Digital Power Analysis Meter) and the speed N (obtained from the Digital RPM Meter) values in Table 3.2.2.

8. Repeat step 7 for other If settings listed in Table 3.2.2.

9. Sequentially turn off the DC Power Supply, Synchronous Machine Exciter, Three-phase Power Supply, 3-P Current Limit Protection Switch Modules and System Transformer.

3.2.6. Results and discussions

1. Fill in the values obtained from the previous section to Table 3.2.2 below:

Table 3.2.2. Measured I_f, E_o, and N

I_f (A)	0	0.02	0.04	0.06	0.08	0.1	0.12	0.14
E_o (V)								
N (rpm)								
I_f (A)	0.16	0.18	0.2	0.22	0.24	0.26	0.28	0.3
E_o (V)								
N (rpm)								

2. Using the results of Table 3.2.2, plot the E_o vs I_f curve. Analyse the relationship between these parameters with relevant theories and equations.

Activity 3.3: Load Characteristic

3.3.1. Aim of activity

After completing this exercise, you should be able to demonstrate the operating characteristic of a three-phase salient pole synchronous generator under load condition.

3.3.2. Equipment required

Table 3.3.1. List of equipment

Qty	Description	Cat. No.
1	DC Permanent-magnet Machine	EM-3330-1A
1	Three-phase Salient Pole Synchronous Machine	EM-3330-3A
1	DC Power Supply Module	EM-3310-1A
1	System Transformer	EM-3340-3B
1	Three-phase Power Supply Module	EM-3310-1E
1	Synchronous Machine Exciter	EM-3310-1C
1	Three-pole Current Limit Protection Switch Module	EM-3310-2A
1	Resistive Load Module	EM-3310-4R
1	Capacity Load Module	EM-3310-4C
1	Inductive Load Module	EM-3310-4L
1	Four-pole Switch Module	EM-3310-2B

1	Digital Power Analysis Meter	EM-3310-3H
2	Digital DCA Meter	EM-3310-3A
2	Digital DCV Meter	EM-3310-3B
1	Digital RPM Meter	EM-3310-3G
1	Coupling	EM-3390-2A
1	Connecting Leads Set	EM-3390-3A
1	Safety Bridging Plugs Set	EM-3390-4A

3.3.3. Schematic

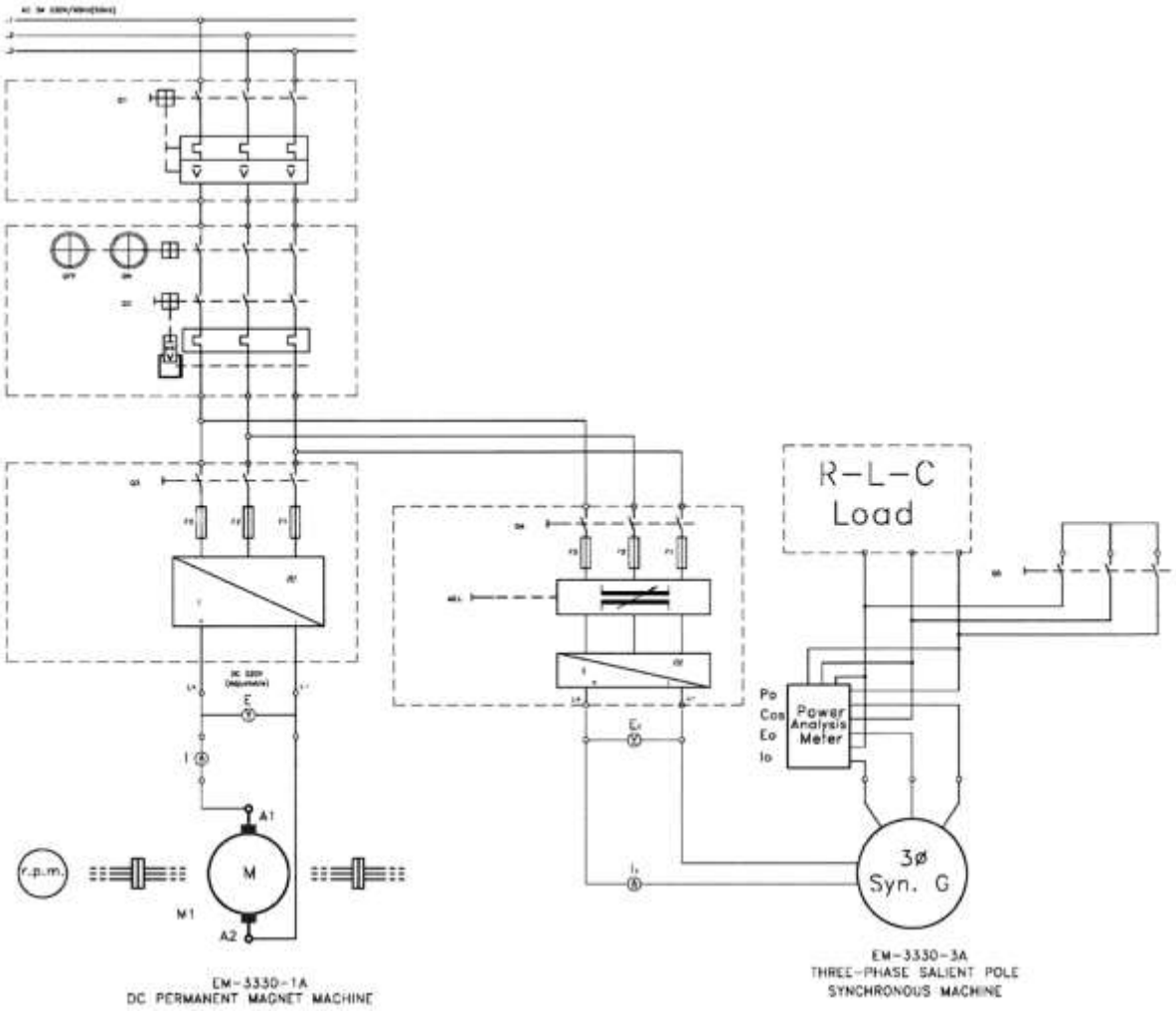


Figure 3.3.1. Circuit diagram for load characteristic test

3.3.4. Connection

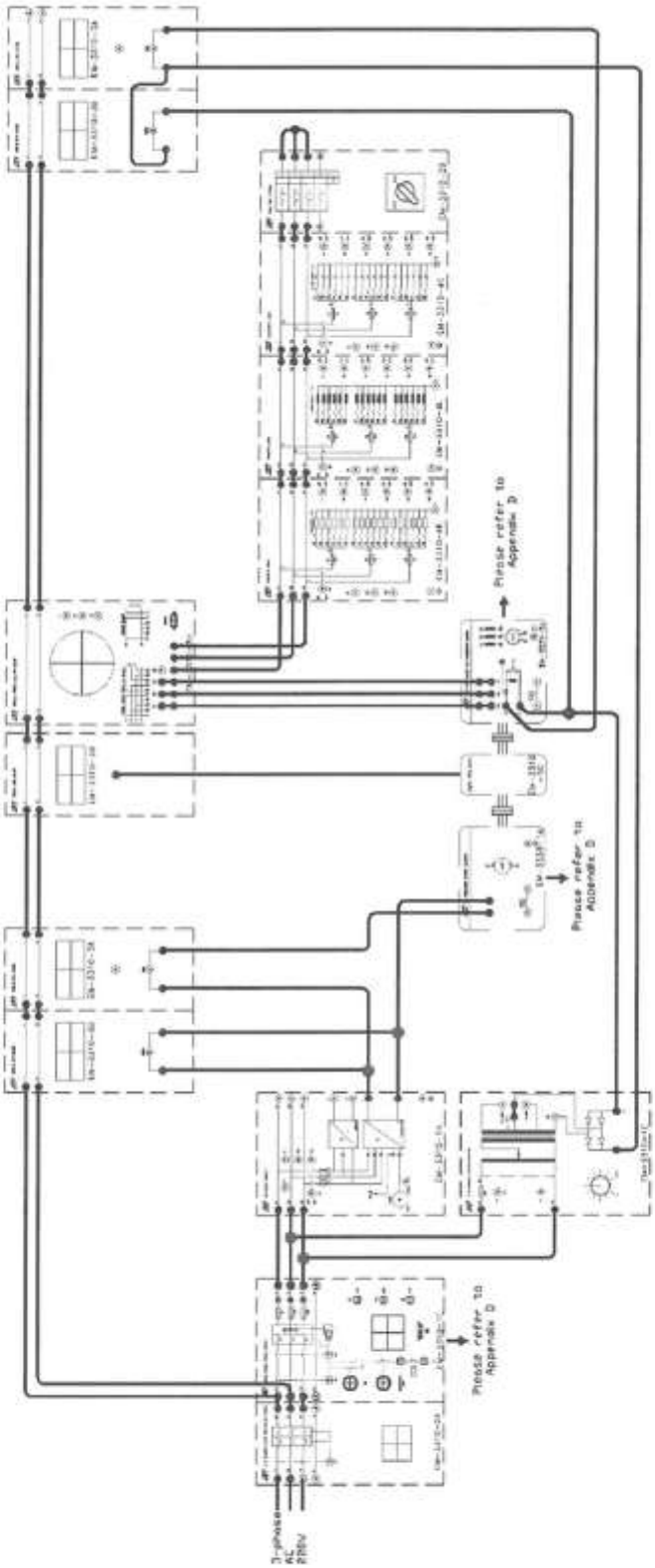


Figure 3.3.2. Connection diagram for load characteristic test

3.3.5. Experimental procedure

1. Place the DC Permanent-magnet Machine, Three-phase Salient Pole Synchronous Machine, and Digital RPM Meter on the Laboratory Table. Mechanically couple the DC Permanent-Magnet (PM) Machine to the Three-phase Salient Pole Synchronous Machine and the Digital RPM Meter using Couplings. Securely lock Machine Bases together using delta screws.

2. Construct the circuit in accordance with the circuit diagram and the connection diagram in Figure 3.3.1 and 3.3.2, respectively. Have the instructor or lab technician to check on your completed circuit. The synchronous generator operates in delta.

3. Set the V_{adj} knob on the DC Power Supply Module to the min. position. Set the voltage control knob on the Synchronous Machine Exciter Module to the 0 position. Set the on-off switch on the Four-Pole Switch module to the OFF position. On the Resistive Load module, set the switches S0 through S4 to ON position.

4. Sequentially turn on the main switches, System Transformer, 3-P Current Limit Protection Switch, Three-phase Power Supply, and DC Power Supply Modules.

5. On the DC Power Supply Module, press the START button and slowly tum the V_{adj} knob to increase the motor voltage until the motor rotates at its rated speed N = 1,500 rpm for 50-Hz power. **Maintain this speed through this exercise.**

6. Turn on the Synchronous Machine Exciter and slowly turn the voltage control knob to set the generator output voltage E_o to 220 V. Record the field current I_f in the S0-S4=ON field of Table 3.3.2. **Note: The I_f value must be maintained in this laboratory excise.**

7. If the rotor of generator is locked by a load, turn off the power immediately. **Note: The motor (prime mover) current must not exceed 130% of the rated current. The generator current must not exceed 130% Of the rated current (*0.8A × 1.3 = 1.04A*).**

8. Maintain the speed (N = 1,500 rpm) and the field current I_f. On the Resistive Load module, set the switches for each of the listed positions in Table 3.3.2. Record the generator output current I_o, generator output voltage E_o, generator output power P_o, and power factor cos θ

(obtained from the Digital Power Analysis Meter), and the generator speed N (obtained from the Digital RPM Meter) values in Table 3.3.2. **The "shorted" position is that the rotor windings are short-circuited by setting the on-off switch on the Four-Pole Switch module to ON position.**

9. Sequentially turn off the Resistive Load, DC Power Supply, Synchronous Machine Exciter, Three-phase Power Supply, 3-P Current Limit Protection Switch Modules and System Transformer.

10. Repeat Steps 3 through 9 for resistive and inductive loads. On the Resistive Load and Inductive Load modules, set the switches for each of the listed positions in Table 3.3.3 and record down the output values.

11. Similarly, repeat Steps 3 through 9 for resistive and capacitive loads. On the Resistive Load and the Capacitive Load modules, set the switches for each of the listed positions in Table 3.3.4 and record down the output values.

12. At the “shorted” position, perform measurement on the generator output current, I_o. Maintain the rated speed of 1500 rpm, and slowly turn the voltage control knob on the Synchronous Machine Exciter Module and set the field current I_f (obtained from the Digital DCA Meter) as listed in Table 3.3.5. The "shorted" position is that the rotor windings are short-circuited by setting the on-off switch on the Four-Pole Switch module to ON position.

13. Sequentially turn off the Resistive Load, DC Power Supply, Synchronous Machine Exciter, Three-phase Power Supply, 3-P Current Limit Protection Switch Modules and System Transformer.

3.3.6 Results and discussions

1. Fill in the values obtained from the previous section to tables below.

2. Using the results of Tables 3.3.2 through 3.3.4, plot the E_o vs I_o curves. Using the results of Table 3.3.5, plot the I_o vs I_f curve.

3. Analyse the relationship between these parameters with relevant theories and equations.

4. Determine the synchronous reactance X_s of this machine.

Table 3.3.2. Measured values of I_f, I_o, E_o, P_o, N and cos θ

	Switch positions on Resistive Load Module							
	S0=OFF	S0-S1=ON	S0-S2=ON	S0-S3=ON	S0-S4=ON	S0-S5=ON	S0-S6=ON	Shorted
I_f (A)								
I_o (A)								
E_o (A)								
cos θ								
P_o (W)								
N (rpm)								

Table 3.3.3. Measured values of I_f, I_o, E_o, P_o, N and cos θ

	Switch positions on Resistive Load and Inductive Load Module							
	S0=OFF	S0-S1=ON	S0-S2=ON	S0-S3=ON	S0-S4=ON	S0-S5=ON	S0-S6=ON	Shorted
I_f (A)								
I_o (A)								
E_o (A)								
cos θ								
P_o (W)								
N (rpm)								

Table 3.3.4. Measured values of I_f, I_o, E_o, P_o, N and cos θ

	Switch positions on Resistive Load and Capacitive Load Module							
	S0=OFF	S0-S1=ON	S0-S2=ON	S0-S3=ON	S0-S4=ON	S0-S5=ON	S0-S6=ON	Shorted
I_f (A)								
I_o (A)								
E_o (A)								
cos θ								
P_o (W)								
N (rpm)								

Table 3.3.5. Measured values of short circuit output current, Io

If (A)	0	0.10	0.20	0.30
Io (A)				
N (rpm)				

Note: Consolidate all the lab activities 3.1, 3.2, and 3.3 into a single report.

APPENDIX 1

MARKING RUBRICS

Component Title	Lab report				Percentage (%)	100	
Criteria	Score and Descriptors					Weight (%)	Marks
	Excellent (5)	Good (4)	Average (3)	Need Improvement (2)	Poor (1)		
Formatting, quality of figures and plots	All sections are presented in a neat and tidy manner with consistent format and style. Complete captions for tables and figures are given and mentioned in the main text. All figures are clear with important labels and comments. Title and the axis of all plots are well labelled, clear, and easy to read.	Most sections are presented in a neat and tidy manner with consistent format and style. Complete captions for tables and figures are given and most are mentioned in the main text. Most figures are clear with labels and comments. Title and the axis of most plots are well labelled and clear.	Some sections are inconsistent in terms of format and style. Most captions for tables and figures are given and mentioned in the main text. Some figures and components in the circuit diagram are not clear and missing important labels and comments. Some title and axis of plots are not well labelled and clear.	Most sections are inconsistent in terms of format and style. Incomplete captions for tables and figures and most are not mentioned in the main text. Most figures and components in the circuit diagram are not clear and missing important labels and comments. Most title and axis of plots are not well labelled and clear.	Messy report and difficult to read with inconsistent format and style. No captions for tables and figures provided. Blurry figures and plots with missing labels provided. Figures and plots are difficult to read and understand.	15	
Introduction and Methodology	Excellently written, concise introduction and methodology with important information that are relevant to the topic.	Well written, concise introduction and methodology with important information that are relevant to the topic.	Averagely written, concise introduction and methodology, consists some irrelevant information.	Averagely written, lengthy introduction and methodology, consists many irrelevant information.	Poorly written, lengthy introduction and methodology, consists many irrelevant information.	15	
Results and discussions	Completed all experiments and data collection. All measured data, calculations, and graphs/plots are correct. Able to evaluate the results with prudent judgment. Analysis is done with excellent description of the relationship involved and explanation with relevant theories.	Completed all experiments and data collection. Most measured data, calculations, and graphs/plots are correct. Results are compared and analysed with good description of the relationship involved and with relevant theories.	Completed most experiments and data collection. Some measured data, calculations, and graphs/plots are incorrect. Results analysis are done with adequate description of the relationship involved.	Completed most experiments, but most measured data, calculations, and graphs/plots are incorrect. Only restatement of the results without commenting on the expected key points.	Experiments not completed. Most measured data, calculations, and graphs/plots are incorrect. Discussion shows little understanding of what the experiment is all about.	50	
Conclusion	The aim of the experiment has been achieved, the key features of the methods used and most important results are summarized.	The aim of the experiment has been achieved, most key features of the methods used and results are summarized.	A conclusion is drawn, but only supported by some experimental evidence.	A conclusion is drawn, but not supported by any experimental evidence.	No sensible conclusion. No comparisons and evidence.	20	
					TOTAL	**100**	

Note to students: Please attach this appendix together with the submission of coursework.